

2. Architecture & Design Documentation

System Overview

- High-level components: API gateway, auth service, application services, background workers, database, cache, object storage, analytics.
- Deployment: containers (Docker) in cloud provider / Kubernetes.

Component / Module Design

- For each module list: responsibilities, interfaces, events produced/consumed, failure modes.

Example: Booking Service

- Responsibilities: create/modify/cancel bookings, enforce business rules.
- Interfaces: `POST /bookings`, internal event `booking.created`.
- Data: booking table (FKs to user, slot).

Sequence Diagrams

- See `diagrams/sequence_diagrams.mmd` for mermaid sequence diagrams.

Data Flow Diagrams

- Describe data ingress (API -> auth -> service -> DB), enrichment, analytics export.

Service Interactions

- Synchronous: REST/HTTP for queries and commands.
- Asynchronous: message queue (e.g., Redis/Stream/Kafka) for events and background jobs.

Design Decisions

- Rationale for microservice vs monolith, data ownership boundaries, eventual consistency tradeoffs.

TODOs:

- Populate component responsibilities and diagrams from the codebase.